Amrita School of Computing

Amritapuri Campus

Department of Computer Science and Engineering

B.Tech Computer Science and Engineering

23CSE498 – Project Phase II

Design and Solution Submission

Project Title Context-Aware Cross-Slice Transformer Framework for Automated Glioma Segmentation Using Multi-Modal MRI

Project Guide Dr. Tintu Vijayan

Team Members

Team No.	Roll Number	Name of Student
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Academic Year: 2026–2027

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1 Sustainable Development Goal (SDG) Mapping

Primary SDG: SDG 3 – Good Health and Well-being

Secondary SDG(s) (Optional): SDG 9 – Industry, Innovation and Infrastructure

Justification (100–150 words):

Gliomas are among the most common primary brain tumors, and accurate segmentation of tumor boundaries from MRI scans is essential for effective surgical planning and radiotherapy targeting. Manual segmentation is currently performed slice-by-slice by radiologists, a process that is time-consuming and subject to inter-observer variability. This project directly supports SDG 3 by developing an automated, consistent, and clinically reliable segmentation framework that can reduce diagnostic delays, support treatment planning accuracy, and ease the workload on radiology teams – particularly valuable in resource-constrained healthcare settings where specialist availability is limited. The project also touches SDG 9, since it contributes a novel AI-driven medical imaging technique (adapting cross-slice attention mechanisms to a new clinical domain) that advances innovation in healthcare infrastructure and diagnostic tooling.

SDG	Goal
1	No Poverty
2	Zero Hunger
3	Good Health and Well-being
4	Quality Education
5	Gender Equality
6	Clean Water and Sanitation
7	Affordable and Clean Energy
8	Decent Work and Economic Growth
9	Industry, Innovation and Infrastructure
10	Reduced Inequalities
11	Sustainable Cities and Communities
12	Responsible Consumption and Production
13	Climate Action
14	Life Below Water
15	Life on Land
16	Peace, Justice and Strong Institutions
17	Partnerships for the Goals

2 Project Milestones and Timeline

Semester	Major Activity / Milestone	Expected Deliverable	Expected Completion	Status
S6	Project domain exploration	Problem domain identified	15 Feb 2026	Completed
S6	Literature review	Initial literature survey	15 Mar 2026	Completed
S6	Baseline implementation	Reference implementation completed	30 Apr 2026	Completed
S7 (Submission 1)	Literature survey	Comprehensive literature survey	15 Jul 2026	Completed
S7 (Submission 1)	Research gap identification	Research gap statement	20 Jul 2026	Completed
S7 (Submission 1)	Problem statement	Finalized objectives	25 Jul 2026	Completed
S7 (Submission 2)	System design	Architecture and workflow	06 Aug 2026	Current
S7	Prototype implementation	Functional prototype	15 Oct 2026	Planned
S7	Module integration	Integrated system	15 Nov 2026	Planned
S7	Testing and validation	Experimental results	15 Dec 2026	Planned
S8	System enhancement	Improved system	31 Jan 2027	Planned
S8	Performance evaluation	Performance analysis	28 Feb 2027	Planned
S8	Research publication	Paper submission	20 Mar 2027	Planned
S8	Dissertation writing	Final report	10 Apr 2027	Planned
S8	Final demonstration	Presentation and viva	20 Apr 2027	Planned

Table 1: Overall Project Timeline

3 System Architecture (If Applicable)

Insert System Architecture Diagram

Replace this placeholder with your figure.

Architecture Description

The proposed system follows a five-stage pipeline. Multi-modal MRI input (T1, T1ce, T2, and FLAIR sequences, taken as a window of five consecutive axial slices) is first passed through a Lightweight Transformer Encoder, which applies patch embedding followed by stacked self-attention blocks to extract multi-scale global features independently for each slice in the window. These per-slice feature representations are then passed to the Cross-Slice Transformer module, the core novelty of this work, which computes attention across the slice dimension so that the prediction for the center slice is informed by its neighboring slices, with an adaptive gating mechanism learning how much each neighboring slice should contribute. The resulting fused representation is passed to the Adaptive Modality Fusion module, which learns attention-based weights across the four MRI modalities rather than treating them as fixed input channels. Finally, the Segmentation Decoder upsamples the fused feature representation back to the original spatial resolution using transposed convolutions, producing a per-pixel class prediction corresponding to background, necrotic/non-enhancing tumor core, peritumoral edema, and enhancing tumor regions. This architecture is designed to capture both fine-grained local detail (via the encoder) and volumetric consistency across the 3D scan (via the cross-slice module), addressing the slice-independence limitation identified in the literature survey.

4 Module Description (If Applicable)

Module	Description
Data Loader & Preprocessing	Reads raw BraTS2020 NIfTI volumes for each patient, applies z-score normalization within the brain region, remaps BraTS label 4 to a dense class index, and extracts sliding windows of five neighboring slices for cross-slice training. Input: patient directories with T1/T1ce/T2/FLAIR/segmentation volumes. Output: normalized slice windows with ground-truth labels.
Lightweight Transformer Encoder	Applies convolutional patch embedding followed by stacked self-attention transformer blocks to each MRI slice independently, extracting multi-scale global feature representations. Input: single slice, 4 modality channels. Output: token feature sequence per slice.
Cross-Slice Transformer	Computes attention across the five-slice window at each spatial token position, with an adaptive gating mechanism that learns how much each neighboring slice should contribute to the center slice's prediction. This is the core novelty module, adapted from Abdikenov et al. (2026). Input: token sequences for all slices in the window. Output: fused token sequence for the center slice.
Adaptive Modality Fusion	Learns attention-based weights across the four MRI modalities (T1, T1ce, T2, FLAIR) so the network can emphasize the most informative modality per spatial location, rather than treating all modalities equally. Input: per-modality token sets. Output: fused token sequence.
Segmentation Decoder	Upsamples the fused token representation back to full input resolution using transposed convolutions, producing per-pixel class logits for the four segmentation classes. Input: fused tokens. Output: pixel-wise tumor segmentation mask.
Loss Computation & Evaluation	Combines Dice loss and weighted cross-entropy loss to handle class imbalance during training, and computes Dice scores separately for the Whole Tumor, Tumor Core, and Enhancing Tumor regions during evaluation, following standard BraTS reporting conventions.

5 System Workflow (If Applicable)

Insert Workflow / Flowchart

Replace this placeholder with your figure.

Workflow Description

The system workflow begins with data ingestion, where raw BraTS2020 patient volumes are read and organized by patient ID, with train/validation splits maintained strictly at the patient level to prevent data leakage across slices of the same patient. Each modality volume is then preprocessed through normalization and label remapping. For each target slice, a window of five neighboring slices is extracted (two before, the target, and two after, with edge-clamping at volume boundaries). Each slice in the window is independently encoded through the Lightweight Transformer Encoder to produce per-slice feature tokens. These are passed to the Cross-Slice Transformer, which fuses information across the slice window using adaptive attention. The fused representation then passes through Adaptive Modality Fusion before being decoded back to full resolution by the Segmentation Decoder. During training, the predicted mask is compared against the expert-annotated ground truth using the combined Dice and cross-entropy loss, and the model weights are updated via backpropagation. During inference, the decoder output is passed through an argmax operation to produce the final tumor segmentation mask, which is evaluated using Dice scores across the Whole Tumor, Tumor Core, and Enhancing Tumor regions.

6 Domain-Specific Design (If Applicable)

Design Component	Details / Reference
Database Schema / ER Diagram	Not applicable – this project does not use a relational database; input data is read directly from BraTS2020 NIfTI volumes on disk.
UML Diagrams	Not applicable – the system is a single ML training/inference pipeline rather than a multi-component software system requiring UML modeling.
Machine Learning Model Architecture	Context-Aware Cross-Slice Transformer: Lightweight Transformer Encoder → Cross-Slice Transformer (adaptive slice fusion) → Adaptive Modality Fusion → Convolutional Segmentation Decoder. See Section 3 for full architecture description.
Dataset Description	BraTS 2020 (Brain Tumor Segmentation Challenge) dataset. Multi-institutional MRI scans with four modalities (T1, T1ce, T2, FLAIR) and expert-annotated ground truth segmentation masks labeling necrotic/non-enhancing tumor core, peritumoral edema, and enhancing tumor.
API Design	Not applicable – no external API is exposed; the project is a research training/evaluation pipeline.
Network Topology	Not applicable.
Security Architecture	Not applicable – BraTS2020 is a de-identified, publicly available research dataset with no patient-identifiable information, so no additional data security layer is required beyond standard access controls on the training environment.
Hardware Block Diagram	Not applicable – see Section 9 for hardware requirements (GPU-based training setup).
Deployment Architecture	Not applicable at this stage – current scope is model training and evaluation; clinical deployment is outside the scope of this academic project.
Other Design Artifacts	Loss function design (combined Dice + weighted cross-entropy) and BraTS-standard evaluation protocol (Whole Tumor / Tumor Core / Enhancing Tumor Dice) – see Section 7.

7 Algorithms / Methodology

Proposed Algorithm / Methodology Description

The proposed methodology trains the Context-Aware Cross-Slice Transformer end-to-end on the BraTS2020 dataset. Patient volumes are split at the patient level into training and validation sets. For each training sample, a window of five consecutive axial slices is extracted and independently encoded via the Lightweight Transformer Encoder. The core contribution, the Cross-Slice Transformer, computes multi-head self-attention across the slice dimension at each spatial token position, allowing the network to model how anatomical structures evolve smoothly between adjacent slices rather than treating each slice as an independent 2D image. A learned adaptive gate assigns a relevance weight to each neighboring slice before the weighted features are fused with the center slice via a residual connection. The fused representation then passes through Adaptive Modality Fusion, which similarly uses learned attention weights to combine the four MRI modalities. The Segmentation Decoder upsamples this representation to produce the final per-pixel class prediction. The model is trained using a combined Dice and weighted cross-entropy loss, optimized with AdamW and a cosine annealing learning rate schedule, with the best checkpoint selected by mean validation Dice score across the Whole Tumor, Tumor Core, and Enhancing Tumor regions.

Pseudocode / Mathematical Formulation (Optional)

Algorithm 1: Cross-Slice Adaptive Fusion

Input: Token features for W neighboring slices, $X \in \mathbb{R}^{W \times N \times D}$
Output: Fused token features for the center slice, $F \in \mathbb{R}^{N \times D}$

1. For each spatial token position $n = 1$ to N:
   (a) $x_n \leftarrow X[:, n, :]$ (features across all W slices at position n)
   (b) $Q, K, V \leftarrow \text{LinearProjections}(x_n)$
   (c) $A \leftarrow \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V$ (attention across the W slices)
   (d) $\text{gate_scores} \leftarrow \text{MLP}(A)$
   (e) $\text{gate_weights} \leftarrow \text{softmax}(\text{gate_scores}, \text{axis} = \text{slice})$
   (f) $\text{fused}_n \leftarrow \sum_w \text{gate_weights}[w] \cdot A[w]$
2. $F \leftarrow [\text{fused}_1, \ldots, \text{fused}_N] + X[\text{center_slice}]$ (residual connection)

3. **return** F

Loss Function:

$$\mathcal{L} = (1 - \lambda)\,\mathcal{L}_{CE} + \lambda\,\mathcal{L}_{Dice}, \qquad \lambda = 0.5$$

Dice Coefficient (per region):

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|}$$

where A is the predicted segmentation region and B is the ground truth region, computed separately for Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

8 Technology Stack (If Applicable)

Category	Version	Technology / Tool
Programming Language	3.10+	Python
Framework	2.1.0+	PyTorch
Frontend	–	Not applicable
Backend	–	Not applicable
Database	–	Not applicable (data read directly from NIfTI files)
Libraries	See notes	einops, nibabel, NumPy, SciPy, scikit-image, Albumentations, TensorBoard, Matplotlib, PyYAML
IDE / Development Environment	–	VS Code / PyCharm / Jupyter Notebook
Version Control	–	Git / GitHub
Cloud / Deployment Platform	–	Google Colab / Kaggle Notebooks (GPU training)
Operating System	–	Ubuntu 20.04+ / Windows 10+
Other Tools	–	Lucid (architecture & workflow diagrams), LaTeX (report)

9 Hardware and Software Requirements

Hardware Requirements

Component	Specification
Processor	Quad-core CPU minimum; 6+ core recommended for parallel data loading
RAM	16 GB minimum; 32 GB recommended
Storage	50 GB minimum free space; 100 GB+ recommended (dataset, checkpoints, logs)
GPU (If Applicable)	NVIDIA GPU with 6 GB+ VRAM minimum (e.g., GTX 1660); 12 GB+ VRAM recommended (e.g., RTX 3060/3080)
Internet Connectivity	Required for dataset download and cloud-based training (if using Colab/Kaggle)
Other Hardware	None

Software Requirements

Software	Version / Details
Operating System	Ubuntu 20.04+ / Windows 10+ / macOS
Programming Language(s)	Python 3.10+
Framework(s)	PyTorch 2.1.0+
Database	Not applicable
IDE / Editor	VS Code / PyCharm / Jupyter Notebook
Libraries / Packages	einops, nibabel, NumPy, SciPy, scikit-image, Albumentations, TensorBoard, Matplotlib, PyYAML
Other Dependencies	CUDA Toolkit 11.8+ (for GPU training)

10 Design Considerations (Non-Functional)

Design Aspect	Description
Scalability	The model architecture is designed to scale across the full BraTS2020 dataset (approximately 370 patient volumes) and can be extended to larger multi-institutional datasets, since the encoder and cross-slice modules operate on fixed-size slice windows independent of total dataset size.
Performance	Evaluated using Dice coefficient across Whole Tumor, Tumor Core, and Enhancing Tumor regions, alongside training/inference time. Lightweight encoder design and windowed cross-slice attention (rather than full 3D attention) keep computational cost manageable on a single GPU.
Security	Not a primary concern at this stage since BraTS2020 is a publicly available, de-identified research dataset; standard access controls on the training environment and codebase (e.g., private GitHub repository) are sufficient.
Reliability	Patient-level train/validation splitting prevents data leakage across slices of the same patient, ensuring evaluation metrics reflect genuine generalization. Model checkpointing ensures training progress is not lost and results are reproducible.
Maintainability	Codebase is organized into clearly separated modules (data loading, model architecture, loss functions, training loop), making it straightforward to modify or extend individual components (e.g., swapping the encoder or adding new fusion strategies) without affecting the rest of the pipeline.
Privacy	BraTS2020 data is fully de-identified and approved for public research use, so no patient privacy concerns apply to this academic project.
Ethical Considerations (If Applicable)	This system is intended as a research tool to support, not replace, radiologist judgment. Any future clinical application would require human oversight, regulatory validation, and awareness that automated predictions may carry biases from the training data distribution.

11 Expected Outcome

The expected outcome of this project is a working implementation of the Context-Aware Cross-Slice Transformer, trained and evaluated on the BraTS2020 dataset, that demonstrates improved volumetric consistency in glioma segmentation compared to slice-independent CNN and Transformer baselines from the literature survey. Concretely, the deliverables are: (1) a trained model achieving competitive or improved Dice scores across the Whole Tumor, Tumor Core, and Enhancing Tumor regions relative to baseline methods, with particular attention to the Enhancing Tumor region, which is typically the hardest to segment accurately; (2) a comparison report benchmarking the proposed model against existing CNN-based and Transformer-based approaches; (3) an ablation study isolating the contribution of the Cross-Slice Transformer and Adaptive Modality Fusion modules individually; and (4) a final project report and prototype suitable for the Phase II prototype implementation milestone (15 Oct 2026, per the project timeline in Section 2).

12 References (Optional)

1. W. Aslam, J. Hussain, M. Z. Aslam, S. Jan, T. B. Riaz, A. Iqbal, M. Arif, and I. Khan, "Enhanced Brain Tumor Segmentation in Medical Imaging Using Multi-Modal Multi-Scale Contextual Aggregation and Attention Fusion," *Scientific Reports*, vol. 15, 2025.

2. P. Song, C. Shen, Z. Fan, F. Yin, Y. Wu, F. Liu, C. Zhao, W. Zeng, X. Xie, G. Wu, B. Yang, Z. Shi, and J. Yu, "An Integrated MRI-Based Diagnostic Framework for Glioma With Incomplete Imaging Sequences and Imperfect Annotations," *npj Precision Oncology*, vol. 9, 2025.

3. R. Li, Y. Liao, Y. Huang, X. Ma, G. Zhao, Y. Wang, and C. Song, "DeepGlioSeg: Advanced Glioma MRI Data Segmentation With Integrated Local-Global Representation Architecture," *Frontiers in Oncology*, vol. 15, Art. no. 1449911, 2025.

4. L. Wen, H. Sun, G. Liang, and Y. Yu, "A Deep Ensemble Learning Framework for Glioma Segmentation and Grading Prediction," *Scientific Reports*, vol. 15, Art. no. 4448, 2025.

5. B. Abdikenov, Z. Kairat, T. Zhaksylyk, Y. Orazayev, B. Ayanbayev, and N. Saidnassim, "Cross-Slice Transformer With Adaptive Slice Fusion for Lymph Node Segmentation in CT Imaging," *IEEE Access*, vol. 14, pp. 98650–98670, 2026.

Faculty Remarks

Guide Remarks

Project Guide

Name: _____

Signature: _____

Project Coordinator

Name: _____

Signature: _____

A Instructions to Students

General Guidelines

- Complete all sections of the report before submission.
- Present technical content in a clear, concise, and professional manner.
- Use original work supported by appropriate references and citations.
- AI-assisted tools may be used only as supporting aids. Students are responsible for ensuring the originality, technical accuracy, and completeness of all submitted content.
- Reports containing unverified, generic, or AI-generated content without substantial student contribution may be rejected.
- All figures, tables, and diagrams must be clearly labeled and referenced in the text.
- Remove all placeholder text before final submission.

Section-wise Guidelines

Project Title Provide a clear and meaningful title that reflects the proposed work.

SDG Mapping Identify the primary SDG and justify its relevance to the proposed project.

Project Timeline Summarize the major milestones, deliverables, and expected completion schedule.

System Architecture
Present the overall system architecture with a brief description of the major components.

Module Description
Describe the functionality of each module and its role in the system.

System Workflow
Illustrate the operational workflow using an appropriate diagram.

Domain-Specific Design
Include only the design artifacts relevant to the project domain.

Proposed Solution

Describe the proposed methodology or novel approach, highlighting its key contribution.

Technology Stack

List the programming languages, frameworks, tools, libraries, and platforms used.

Requirements Specify the hardware and software required for development and deployment.

Design Considerations

Discuss relevant aspects such as performance, scalability, security, maintainability, and reliability.

Expected Outcome

Summarize the expected deliverables and anticipated project outcomes.

References Cite all referenced literature using IEEE style.

Submission Checklist

- All required sections are completed.
- Figures, tables, and references are properly numbered and cited.
- Grammar and formatting have been reviewed.
- The report has been verified and approved by the project guide prior to submission.